

XIANGYU WANG

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EDUCATION

Bielefeld University (Germany)

Oct. 2022 - Present

- Ph.D. student in Computer Science, supervised by Prof. Dr.-Ing. Yaochu Jin, MAE, FIEEE, and Prof. Dr.-Ing. Ulrich Rückert
- Research Interests: Multi-objective evolutionary computation, neural combination optimization

China University of Petroleum (East China)

2016 - 2020 - 2022

- B.S. and M.S. in Mathematics, supervised by Prof. Jian Wang.
- Curriculum: Mathematical analysis, advanced algebra, optimization

SKILL

- Programming languages: Python (PyTorch), MATLAB.

PUBLICATIONS (IN JOURNAL)

1. **X. Wang** and Y. Jin, “A size-imbalanced dual population with complementary search for large-scale multi-modal multi-objective optimization”, *accepted, in IEEE Transactions on Evolutionary Computation*, 2025. (IF: 12) [Paper link](#)
2. T. Zheng, J. Liu, Y. Jin, **X. Wang**, Y. Liu, “Large-scale multimodal multiobjective optimization based on multiview diversity enhanced mechanism”, *accepted, IEEE Transactions on Evolutionary Computation*, 2025. (IF: 12) [Paper link](#)
3. **X. Wang**, R. Cheng, and Y. Jin, “Sparse large-scale multi-objective optimization by identifying non-zero decision variables”, in *IEEE Transactions on Systems, Man, and Cybernetics: System*, vol. 54, no. 10, 2024. (IF: 8.7) [Paper link](#)
4. **X. Wang**, X. Yan and Y. Jin, “A graph neural network with negative message passing and uniformity maximization for graph coloring”, in *Complex & Intelligent Systems*, vol. 10, 2024. (IF: 5.1) [Paper link](#)
5. **X. Wang**, K. Zhang, J. Wang, Y. Jin, “An enhanced competitive swarm optimizer with strongly convex sparse operator for large-scale multi-objective optimization”, in *IEEE Transactions on Evolutionary Computation*, vol. 26, no. 5, 2022. (IF:12) [Paper link](#)
6. **X. Wang**, B. Zhang, J. Wang, K. Zhang, Y. Jin, “A cluster-based competitive particle swarm optimizer with a sparse truncation operator for multi-objective optimization”, in *Swarm and Evolutionary Computation*, vol. 71, 2022. (IF: 8.5) [Paper link](#)
7. **X. Wang**, B. Mickiewicz, *et al.*, “Comparison of two automated targeted metabolomics programs to manual profiling by an experienced spectroscopist for 1H-NMR spectra”, in *Metabolites*, vol. 12, no. 3, 2022. (IF: 3.7) [Paper link](#)
8. **X. Wang**, J. Wang, K. Zhang, F. Lin, Q. Chang, “Convergence and objective functions of noise-injected multilayer perceptrons with hidden multipliers”, in *Neurocomputing*, vol. 452, 2021. (IF: 6.5) [Paper link](#)
9. B. Liu, H. Lin, F. Li, **X. Wang**, “Blow-up analyses in reaction–diffusion equations with nonlinear nonlocal boundary flux,” in *Zeitschrift für angewandte Mathematik und Physik*, vol. 70, 2019. (IF: 1.7) [Paper link](#)

PUBLICATIONS (IN CONFERENCE)

1. **X. Wang**, X. Wang, Y. Jin, and U. Rückert, “A graph neural network assisted evolutionary algorithm for expensive multi-objective optimization”, WCCI, 2024, Yokohama, Japan. [Paper link](#)
2. **X. Wang**, T. Zheng and Y. Jin, “Adaptive merging and coordinated offspring generation in multi-population evolutionary multi-modal multi-objective optimization”, DOCS, 2023, Tianjin, China. [Paper link](#)
3. **X. Wang**, and Y. Jin, “An ant colony algorithm assisted by graph neural networks for solving vehicle routing problems”, GECCO workshop, 2023, Lisbon, Portugal. [Paper link](#)
4. J. Li, **X. Wang**, *et al.*, “Sparse broad learning system via a novel competitive swarm optimizer”, IAEAC, 2022, Beijing, China. [Paper link](#)

5. Z. Rong, X. Gong, **X. Wang**, *et al.*, “A slime mold fractional-order ant colony optimization algorithm for travelling salesman problems”, ICSI, 2021, Qingdao, China. [Paper link](#)

PAPERS UNDER REVIEW

1. **X. Wang** and Y. Jin, “Heterophily-aware graph contrastive learning with LLM-assisted evolutionary structural augmentations for graph coloring”, *under review, IEEE Transactions on Neural Networks and Learning Systems*. (IF: 8.9)
2. **X. Wang**, P. Yue, Y. Jin, “Efficient structure-conditioned diffusion-based optimization for featureless graph coloring”, *under review, IEEE Transactions on Neural Networks and Learning Systems*. (IF: 8.9)
3. **X. Wang** and Y. Jin, “Capturing structural information in graph coloring with an enhanced recurrent graph neural network”, *under review, IEEE Transactions on Emerging Topics in Computational Intelligence*. (IF: 6.5)

CONFERENCE ORGANIZATION

Workshop Organizer - WCCI 2026

Jun. 2026

Will organize the workshop in 2026 IEEE World Congress on Computational Intelligence (WCCI 2026) in Netherland, titled ‘*Generative and Adaptive Surrogate Models for Evolutionary Algorithms*’. [Website](#)

Workshop Organizer - WCCI 2024

Jun. 2024 - Jul. 2024

Organized the workshop in 2024 IEEE World Congress on Computational Intelligence (WCCI 2024) in Japan, titled ‘*Privacy-Preserving and Fairness-Aware Optimization*’. [Website](#)

Special Session Organizer - WCCI 2024

Jun. 2024 - Jul. 2024

Organized the Special Session in 2024 IEEE World Congress on Computational Intelligence (WCCI 2024) in Japan, titled ‘*Machine Learning assisted Heuristics for Combinatorial Optimization*’ as a session chair. [Website](#)

Workshop Organizer - AJCAI 2023

Nov. 2023

Organized the workshop in the Australasian Joint Conference on Artificial Intelligence (AJCAI 2023) titled ‘*Machine Learning for Data-driven Optimization*’. [Website](#)

RESEARCH EXPERIENCE, FUNDING, AND HONORS

Doctoral Researcher, Bielefeld University, Germany

Oct. 2022 - Present

Graph Neural Optimization for Graph Coloring and Beyond

✓ Funding: Supported by the Bielefeld Young Researchers’ Fund

✓ Funding: Supported by the Alexander von Humboldt Professorship for Artificial Intelligence awarded to Prof. Yaochu Jin. [Website](#)

Master’s Research, China University of Petroleum, China

Sep. 2020 - Jun. 2022

Evolutionary Algorithms for Large-Scale Sparse Multi-Objective Optimization Problems

✓ Funding: Supported by the UPC Graduate Innovation Project in 2021 (No. YCX2021146, Project Leader).

★ Honors: Received the Excellent Graduation Thesis Award from UPC and Shandong Province.

Visiting Undergraduate Student, University of Calgary, Canada

Jan. 2020 - Jul. 2020

Statistical Methods based on Metabolites

✓ Funding: Supported by China Scholarship Council (No. 201906450115, A1).

★ Honors: Received the Excellent Undergraduate Thesis Award from UPC.

Bachelor’s Research, China University of Petroleum, China

Sep. 2018 - Jun. 2020

Simultaneous Non Simultaneous Singularities of Solutions of Nonlinear Partial Differential Equations

✓ Funding: Supported by China College Innovation and Entrepreneurship Program.

★ Honors: Project Leader, National Excellent Project, No. 201910425120.